Decomposing visual masking into stimulus type, masking procedure, and the general visual factor: Only the general visual factor relates to cognitive performance

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Introduction

One of the most striking findings in individual-differences research is that performance on simple perceptual tasks predict performance on high-level cognitive task such as working memory and fluid intelligence task (Jastrzębski, Kroczek, & Chuderski, 2021; Robison, Campbell, Garner, Sibley, & Coyne, 2026; Tsukahara, Harrison, Draheim, Martin, & Engle, 2020). Perhaps the best documented link between perceptual ability and higher-order cognition comes from the inspection time (IT) task. This finding was first demonstrated by Galton (1883). The classic version of the task is shown in Figure 1. Here, individuals briefly view a simple figure and make a perceptual judgment. The brief display of the simple figure is followed by a pattern mask such that the visual information is available for a limited time. Consequently, accuracy is not at ceiling (Vickers, Nettelbeck, & Willson, 1972). Individuals with higher accuracy on these inspection-time tasks tend to score higher on tests of intelligence (Deary & Stough, 1996; Nettelbeck & Lally, 1976), with correlations of about .3 (Grudnik & Kranzler, 2001; Kranzler & Jensen, 1989).

Why should accurate judgement about brief displays be related to high performance in working memory and IQ tasks? The standard answer is based on *processing speed*. People who do well on the IT task are good at extracting visual information quickly. They have fast visual processing speed. The claim then is that processing speed is not domain specific but is a general phenomenal that affects many cognitive processes (Jensen, 2006; Salthouse, 1996; Vernon, 1983). Individuals with fast processing speed do well in working memory and IQ tasks because it is general.

The processing speed account of the correlation makes sense. Yet, we worry here about the claim that inspection time indexes processing speed *per se*. Certainly, it is plausible that individual differences in inspection time tasks reflect processing speed, but,

on the other hand, they may reflect factors that do not have a temporal basis. For example, the task also requires individuals to maintain good acuity and filtering. In the modern literature, perceptual abilities that are not time limited also correlate with intelligence and working memory (Deary, Bell, Bell, Campbell, & Fazal, 2004; Mikellidou, Lambrou, Georgiou, & Avraamides, 2023; Mosing, Pedersen, Madison, & Ullén, 2014; Tsukahara et al., 2020). A good example of this correlation comes from Tsukahara et al. (2020) who measured how well individuals could discriminate between the loudness of tones, the pitch of tones, the length of lines, and the area of circles. These discriminate abilities correlated between xx and yy with progressive matrix reasoning tasks. For example, Tsukahara et al. (2020) report a correlations between xx and yy between sensory discrimination and progressive-matrix reasoning. These findings do not rule out a processing speed explanation *per se* as these tasks may require speediness in maintaining representations across different fixations and attentional moments for comparison.

Jastrzębski et al. (2021) run a series of IT tasks, a series of sensory discrimination tasks, and a number of working memory and intelligence tasks. They find almost a perfect triangle of correlations. The latent variable *inspection-time* ability correlates about .65 with the latent variable blah blah blah, cant learn much.

Our goal is to put spatial and temporal performance in contrast and ask which form better correlates to higher-order cognition. For the purposes herein, we distinguish between *fast eyes* and *sharp eyes*. Individuals with fast eyes process visual information quickly; people with sharp eyes process spatial information well in the presence of visual noise. If the fast-eyes account is correct, the relation between IT and cognitive performance arises because the backward mask limits the time available to process the target. Remove this time limit, and the relation should disappear. On the other hand, if the sharp-eye account is correct, then relation reflects the ability to see under impoverished visual conditions. Improve the visual conditions, and the relations should disappear.

These diverging predictions may be tested in a highly controlled environment where only the type of masking differ. In the *backwards-masking* versions of the tasks, the stimuli were relatively clear and the key manipulation was the time between stimulus and a following patterned backwards mask. In the *spatial-masking* versions, the stimulus was presented in stimulus-tuned noise, and the stimulus was displayed until response. In the backwards masking version, the duration of information is manipulated and individual differences rely more on processing speed (fast eyes) than acuity (sharp eyes). In the spatial masking version, the amount of noise is manipulated and individuals differences reply more on acuity than processing speed. Key in this setup is that all other aspects of the stimulus are fairly constant across backward-masking and spatial-masking versions. Thus, the experiment has the potential to discriminate between the fast-eyes and sharp-eyes hypotheses. If the limit is fast eyes, then backwards-masking versions should correlate with higher-order tasks better than spatial-masking versions do. Conversely, if the limit is sharp eyes, the reverse should hold.

We construct six different visual perception tasks by crossing three stimulus types with the two mask types. The two mask types are whether the stimulus is degraded by backwards masking or by spatial masking. The three stimulus types are whether the target is a "pi" stimulus which is standard in imspection time tasks, a letter stimulus, or a gabor grating. The pi stimulus consists of two vertical segments connected by an upper horizontal segment, and the individual judges whether the left or right leg is longer. Letter stimuli are just these, and the individual judges which letter is presented. Gabor stimuli have an orientation that is either -45 degrees from the vertical (left tilting) or +45 degrees from the vertical (right tiling)/ Individuals judge whether the gabor orientation is left or right tilting. Figures X blah blah

The expected data patters are as follows: if the fast-eyes hypothesis holds, then the three backwards-masking tasks should cluster well and the combined latent variable should correlate well with higher-order tasks. Conversely, if the sharp-eyes hypothesis holds, then

the the three spatial-masking tasks should cluster well and the combined letat variable should correlate well with higher-order tasks. Yet, there are data patterns that do not accord to either hypothesis. Perhaps, there is no special concept of fast or sharp eyes. An alternative is that some people have better vision that others, and that those with better vision tend to do better on the higher-order cognitive functions. We call this the *good-eyes* hypothesis.

Given the large literature on the correlation between inspection time and IQ, we ran this experiment to confirm that processing speed was crucial. We fully expected the backwards-making tasks to cluster and the shared variance among them to drive the correlation between perception and higher-order cognition. We were quite wrong in these expetations. To foreshadow, we find that the distinction between spatial and backwards masking accounts for very little of the variance across individuals. A factor that accounts for more variance is stimulus type. It seems that people vary more across gabors, letters, and pi stimuli than they do across spatial and backwards masks. Moreover, it seems that what is shared across the stimulus types that drives the correlation with higher-order cognition. And this fact provides the best evidence that neither spatial and nor backwards masking are privileged in relating to higher-order cognitive functions. It seems the good-eyes hypothesis, our add on hypothesis for completeness, is best describes the data.

Jeff Stops Here.

There is a feature of modern inquiries that work against answering the good-eyes vs. fast-eyes questions. To our knowledge previous researchers use different stimuli to measure processing speed and spatial acuity. Processing speed is almost always measured with a set of stimuli that we call π stimuli. The stimulus, shown in Figure 2, consists of two vertical segments connected on top by a horizontal segment; taken together the stimulus is a rough approximation of the Greek letter π. The individual judges which of the two vertical segments is longer. Spatial acuity, in contrast, may be measured as how

well tones of various loudness may be discriminated or circles of various areas may be discriminated. And, although line-length discrimination is used, it is never set up in a π-stimulus configuration. We think this incompatibility is a mistake, and it would be best to have carefully matched spatial and temporal abilities tasks.

A recent study of the relationship among spatial acuity, processing speed, and high-level cognitive performance from Jastrzębski et al. (2021) illustrates this problem. They measured processing speed with choice reaction time (CRT) tasks using letters, digits, and arrows, and spatial acuity with entirely different stimuli (e.g., circles, squares, rectangles, dot arrays, and tones). They found that the two constructs were separable (r = .34–.62). Because the processing speed and acuity measures differed in both stimulus and task format, however, this separation may reflect stimulus-specific variance rather than a difference between the underlying abilities. Their own data are consistent with this concern: even within spatial acuity, the visual and auditory factors could not be merged into a single cluster (r = .59). Thus, whether processing speed and spatial acuity are truly distinct remains an open question that requires matched stimuli to resolve.

To address the stimulus confound in previous studies and to ask which source of variance relates to cognition, we crossed three stimuli (a Gabor patch, a letter, and the Pi stimulus) with the two masking procedures, using an identical mask for each stimulus under backward and spatial masking. In this fully crossed design, variance associated with the masking procedure can be separated from variance associated with the stimulus (Campbell & Fiske, 1959). If fast eyes and good eyes are distinct abilities, tasks that share a masking procedure should correlate more strongly than tasks that share a stimulus; if individual differences are organized by stimulus, the reverse should hold. The same individuals completed three complex span tasks of working memory (Conway et al., 2005; Pahor, Seitz, & Jaeggi, 2022; Unsworth, Heitz, Schrock, & Engle, 2005) and a matrix reasoning test of fluid intelligence (Pahor, Stavropoulos, Jaeggi, & Seitz, 2019), which together served as our measure of cognitive performance. We estimated thresholds with a

hierarchical psychometric model and decomposed their variance into a general factor, masking-specific factors, and stimulus-specific factors, which allowed us to ask which of these sources relates to cognitive performance.

Method

individuals

Eighty-three undergraduate students at the University of California, Irvine, participated in exchange for $50 in total compensation. Four individuals withdrew before completing all sessions, leaving a final sample of 79. All individuals reported normal or corrected-to-normal vision and provided informed consent. The study was approved by the Institutional Review Board of the University of California, Irvine (Study ID: STUDY00000064).

Apparatus

individuals were tested individually in a dimly lit cubicle, seated approximately 100 cm from a 24-inch BenQ LED gaming monitor (1920 × 1080 pixels, 165 Hz refresh rate, 6.06 ms per frame). Stimuli were generated with PsychoPy3 (Peirce et al., 2019) in Python 3.8, running on a PC with Lubuntu Linux 22.04.

Tasks Overview

To assess the sensory discrimination ability and processing speed, the experiment contained six visual masking tasks: three employing backward masking (see Figure 1) and three employing spatial masking (see Figure 2). Both sets of tasks used the same stimulus classes (a Gabor stimulus, a letter stimulus, and a Pi stimulus) and mask classes. The critical difference was the masking procedure: in the backward masking tasks, the mask

followed the stimulus after a variable stimulus–onset asynchrony (SOA), whereas in the spatial masking the stimulus and mask were presented spatially.

To assess cognitive performance, individuals completed three complex span tasks of working memory (OSPAN, CSPAN, and SSPAN) and one matrix reasoning test of fluid intelligence (UCMRT).

Sensory discrimination tasks

Gabor Stimuli. individuals completed a two-alternative forced-choice orientation discrimination task. On each trial, a fixation cross was presented for 485 ms, followed by a centrally presented, random-phase Gabor patch (±45° orientation) embedded in white noise, which remained on the screen for another 485 ms. individuals indicated whether the grating was tilted up-left or up-right by pressing the keyboard. Task difficulty was adaptively adjusted by varying stimulus contrast.

Letter Stimuli. individuals performed a forced-choice letter identification task using the keys A–L (second keyboard row). Each trial began with a 485 ms fixation cross, followed by a centrally presented target letter (one of A, S, D, F, G, H, J, K, L) displayed for 485 ms. The letter was embedded in a frequency-matched letter noise mask constructed as follows: We constructed the mask by first computing the pixelwise average of the nine possible letters and then perturbing its Fourier spectrum with complex Gaussian noise. After inverse transformation and normalization, this yielded a structured noise pattern with the same overall spatial frequency profile as the average letter image. On each trial, the target letter was linearly blended with this mask, and individuals identified the letter using the corresponding keyboard key. Task difficulty was adaptively adjusted by varying the target weight in the letter–noise blend.

Pi Stimuli. individuals performed a forced-choice judgment on a Π-shaped (Pi) target, indicating which vertical "leg" was longer (left vs. right). Each trial began with a

485 ms fixation cross, followed by a centrally presented Π stimulus for 485 ms. The mask was composed of random oblique line segments arranged to form a Π-like clutter. Short line elements (randomized in length and orientation within ±10°–45°) were placed in two vertical columns aligned with the legs of the Π and around the crossbar region. Equal numbers of lines were drawn on each side, although their random placement could introduce local asymmetries. On each trial, the target Π was superimposed on this cluttered mask, and individuals indicated which leg appeared longer using the keyboard. Task difficulty was adaptively controlled by varying the leg-length difference.

Processing speed tasks

Gabor Stimuli. The task matched the spatial Gabor-orientation discrimination task. Each trial began with a 485 ms fixation cross, followed by a centrally presented, random-phase Gabor patch (±45° orientation). After a variable SOA, the target was replaced by a dynamic sequence of five white noise masks (48.5 ms each). individuals indicated whether the grating was tilted up-left or up-right by pressing the keyboard. Task difficulty was adaptively adjusted by SOA.

Letter Stimuli. The task matched the spatial letter-identification condition. Each trial began with a 485 ms fixation cross, followed by a centrally presented target letter (one of A, S, D, F, G, H, J, K, L). After a variable SOA, the target was replaced by a dynamic sequence of five frequency-matched letter noise masks (48.5 ms each). individuals identified the letter using the corresponding keyboard key. Task difficulty was adaptively adjusted by varying SOA.

Pi Stimuli. The task matched the spatial Π-leg discrimination condition. Each trial began with a 485 ms fixation cross, followed by a centrally presented Π stimulus. After a variable SOA, the target was replaced by a dynamic sequence of five clutter masks composed of randomly oriented oblique line segments (48.5 ms each). individuals indicated

which leg appeared longer using the keyboard. Task difficulty was adaptively controlled by varying SOA.

Cognitive tasks

OSPAN. individuals completed an automated OSPAN task (Unsworth et al., 2005) combining simple math operations with working memory recall for letter sequence. Each set contained 3–7 letters, with each set size presented three times, yielding 15 memory sets in total. Before the main task, individuals completed three practice sets of size three to familiarize themselves with the procedure. In the main task, set sizes 3–7 were each presented three times in randomized order (15 sets total). Within each set, trials alternated between (i) a math operation screen (e.g., "6 + 2 = ?"), after which the individual clicked to continue, and (ii) a proposed answer screen showing a single digit; individuals judged whether the digit correctly solved the operation by clicking true/false. A single uppercase letter was then shown for 800 ms, and the sequence continued until the set ended. At recall, individuals clicked the letters in serial order from a matrix. For each set, individuals earned points equal to the set size only if (a) all math operations were solved within the individual's mean solving time in the practice trials plus 2.5 SD, (b) no math errors occurred for that set, and (c) all letters were recalled in the correct order. Sets failing any criterion scored 0. The final OSPAN score was the sum of points across all sets.

CSPAN. individuals completed a CSPAN task (Conway et al., 2005) combining serial counting with working memory recall for number sequence. Each display contained a set of target items (green circles) intermixed with distractors (yellow circles). individuals counted the number of targets in each display, after which the individual clicked to continue to the next display. At recall, individuals were prompted to recall the sequence of counts (e.g., "3–5–2") in the exact order they were encountered. Responses were made by clicking the digits from a matrix. The procedure began with three practice sets of size 2. The main task then presented set sizes 3–7, each three times in randomized order (15 sets total). A

set contributed points equal to its size only if the final recall sequence was accurate in both values and order. The final CSPAN score was the sum of points across all sets.

SSPAN. individuals completed a SSPAN task adapted from the Corsi block-tapping paradigm to assess both storage and processing aspects of spatial working memory (Pahor et al., 2022). On each trial, cartoon gophers appeared sequentially in one of 12 possible screen locations, each presented for 1,500 ms with a 500 ms inter-stimulus interval. individuals were required to reproduce the sequence by tapping the corresponding locations in the exact order. Between each gopher presentation individuals performed a secondary sorting task, dragging an item to the left or right, thereby imposing concurrent processing demands and limiting reliance on mnemonic strategies. Sequence length ranged adaptively from 2 to 10 items, increasing or decreasing in difficulty depending on accuracy. The final SSPAN score was the total successfully recalled items across all sets.

UCMRT. individuals completed the UCMRT (Pahor et al., 2019), a measure of fluid intelligence modeled after Raven's Progressive Matrices. Each trial presented a 3×3 matrix of abstract visual patterns with the bottom-right cell missing, and individuals selected the option that best completed the matrix from eight alternatives displayed below. UCMRT was chosen because it was especially developed for young adults with above-average performance on matrix reasoning tasks, providing a more sensitive and efficient assessment of individual differences in fluid intelligence in this population. The task consisted of 23 problems of varying difficulty, including 2 two-relation items, 3 three-relation items with a single transformation, 6 three-relation items with two transformations, 6 three-relation items with three transformations, and 6 logic items. Performance was scored as accuracy, defined by the number of correctly solved matrices.

Procedure

The experiment consisted of three sessions of approximately one hour each, conducted on separate days within one week. Each session comprised three blocks, with short breaks permitted between blocks. The first and third blocks were visual masking blocks, and the middle block contained the cognitive tasks for that session: OSPAN on the first day, CSPAN on the second day, SSPAN task and the UCMRT on the third day.

Within each visual masking block, individuals completed all six visual masking tasks in random order. Each task consisted of 50 trials, on which a two-down–one-up and a three-down–one-up staircase strictly alternated, yielding 25 trials per staircase. Because the two staircases converge on different performance levels, approximately 70.7% and 79.4% correct, respectively (Levitt, 1971), alternating them allowed the psychometric function to be estimated more precisely. In total, each individual completed 300 trials per task (6 blocks $\times$ 50 trials). The first masking block served as practice and was excluded from analysis, leaving 250 trials per task (125 per staircase) and 1,500 masking trials per individual for analysis.

Data exclusion

Data were screened separately for each individual and each masking task. A individual's data on a given task were excluded from further analysis if either of two criteria was met. First, the individual's raw threshold, computed as the mean stimulus level across the 250 analyzed trials, deviated by more than four standard deviations from the group mean threshold for that task. Second, the observed accuracy on either staircase deviated by more than 5 percentage points from its target accuracy (70.7% for the two-down–one-up staircase and 79.4% for the three-down–one-up staircase), indicating that the staircase had not converged as intended. In total, 14 task datasets from 10 individuals were excluded. These individuals' remaining tasks were retained in the analysis.

Hierarchical psychometric model

Individual thresholds on the masking tasks and the correlations among them were estimated with a hierarchical psychometric model (Rouder & Mehrvarz, 2026). The model estimates individual thresholds and their correlations spatially from trial-level data, thereby accounting for trial-to-trial noise and avoiding the attenuation of correlations caused by measurement error.

Let $i = 1, \ldots, I$ denote individuals, $j \in \{1, 2, 3, 4, 5, 6\}$ denote six masking tasks, and $k = 1, \ldots, K$ denote trials. Let y_{ijk} denote the binary correctness outcome on the k-th trial of individual i in masking task j. We assume

$$y_{ijk} \sim \text{Bernoulli}(F(s_{ijk}; \theta_{ij}, \beta_j)) \,,$$

where the success probability is given by a task-specific psychometric function, modeled as a cumulative normal function of log stimulus strength with a lower asymptote at chance:

$$F(s_{ijk}; \theta_{ij}, \beta_j) = b_j + (1 - b_j)\, \Phi(\beta_j\, (s_{ijk} - \theta_{ij})) \,.$$

Here, $\Phi(\cdot)$ denotes the standard normal cumulative distribution function, and b_j is the guessing rate for task j, fixed at .5 for the two-alternative Gabor and Pi tasks and at 1/9 for the nine-alternative Letter tasks. The variable s_{ijk} denotes the stimulus strength on trial k for individual i on task j, log-transformed and then standardized within each task. In the backward masking tasks, stimulus strength was the SOA; in the spatial masking tasks, it was the Gabor contrast, the target weight in the letter–noise blend, and the leg-length difference of the Pi stimulus, respectively. The parameter θ_{ij} denotes the latent threshold of individual i on task j, expressed on the same scale as s_{ijk}, and β_j denotes the slope of the psychometric function for task j.

At the latent level, each individual is characterized by a vector of ten person-level parameters: the six latent thresholds from the masking tasks and the standardized scores on the four cognitive tasks,

$$\boldsymbol{\theta}_i = (\theta_{i1}, \ldots, \theta_{i6}, c_{i1}, \ldots, c_{i4})^\top ,$$

where c_{im} denotes the standardized score of individual i on cognitive task m. These ten parameters are assumed to follow a multivariate normal distribution,

$$\boldsymbol{\theta}_i \sim \mathcal{N}_{10}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) ,$$

where $\boldsymbol{\mu}$ captures the mean of each task and $\boldsymbol{\Sigma}$ captures the covariance among tasks. The covariance matrix is given a scaled inverse-Wishart (SIW) prior (Huang & Wand, 2013) which is robust to differences in the measurement scales of the tasks and is computationally efficient to estimate (Yang & Rouder, 2026). Full details of the model specification are provided in Appendix A.

Results

Individual differences and their correlational structure

Figure 3 summarizes the model-estimated scores for the six masking tasks and the four cognitive tasks. All scores were standardized before plotting. The distributions on the diagonal were unimodal and roughly symmetric, with substantial spread across individuals. Thus, all ten tasks captured meaningful individual differences. The lower triangle of Figure 3 shows the model-estimated correlations among the ten tasks. All 45 correlations were positive with a single exception between the backward Letter task and SSPAN (r = -.02). The mean correlation was .39 among the six masking tasks, .35 among the four cognitive tasks, and .26 between the masking tasks and the cognitive tasks.

It is worth noting that the three same-stimulus pairs, in which the same stimulus was tested under backward and spatial masking, showed the three highest correlations among the masking tasks (mean $r = .52$). By comparison, the mean correlation was .40 between tasks that shared a masking procedure but differed in stimulus, and .31 between tasks that differed in both. This pattern suggests that stimulus type plays a substantial role in individual differences in masking performance. More direct evidence comes from hierarchical clustering, an unsupervised method that groups tasks according to their similarity without imposing any a priori structure (see Figure 4). The first three merges each joined the two masking procedures of the same stimulus; only after these pairs had formed did they combine with one another. By comparison, neither the three backward masking tasks nor the three spatial masking tasks formed a cluster of their own. The four cognitive tasks formed a separate cluster, which joined the masking tasks only at the final step. Thus, individual differences in masking performance were organized primarily by stimulus rather than by masking procedure.

Structural decomposition of masking performance

Hierarchical clustering showed that the structure among the masking tasks is best described by grouping by stimulus type. The next question is whether stimulus type or masking procedure contributes to cognitive performance. These two questions are different because the covariance that determines how tasks cluster need not be the covariance that predicts cognitive performance. Tasks sharing a stimulus may cluster together because of stimulus-specific variance, while their relation to cognitive performance may instead be carried by other sources of variance, for example, variance from the masking procedure or variance from a general visual factor.

To answer this question, we retained the data level of the hierarchical psychometric model and replaced the covariance matrix $\mathbf{\Sigma}$ at the latent level with a structural equation model. For the six masking tasks, the thresholds θ_{ij} remain linked to the trial-level data

through the psychometric function $F(s_{ijk}; \theta_{ij}, \beta_j)$; what changes is how the person-level vector $\boldsymbol{\theta}_i$ is generated. In the new latent structure, $\boldsymbol{\theta}_i$ is expressed as a function of seven latent factors,

$$\boldsymbol{\eta}_i = \left(\eta_i^V, \eta_i^{\text{Bwd}}, \eta_i^{\text{Sim}}, \eta_i^{\text{Let}}, \eta_i^{\text{Gab}}, \eta_i^{\text{Pi}}, \eta_i^C\right)^\top,$$

where η_i^V is a general visual factor shared by all six masking tasks; η_i^{Bwd} and η_i^{Sim} are masking factors, each shared by the three tasks that use the same masking procedure; η_i^{Let}, η_i^{Gab}, and η_i^{Pi} are stimulus factors, each shared by the two tasks that use the same stimulus; and η_i^C is a cognitive factor measured by the four cognitive tasks. We refer to the six factors other than η_i^C collectively as non-cognitive factors.

The measurement model relates the ten person-level parameters to these factors,

$$\boldsymbol{\theta}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda}\,\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}_{10}(\mathbf{0}, \boldsymbol{\Delta}), \qquad \boldsymbol{\Delta} = \text{diag}\left(\delta_1^2, \ldots, \delta_{10}^2\right),$$

where $\boldsymbol{\Lambda}$ is a 10×7 matrix of factor loadings and δ_j^2 is the variance unique to task j. With the rows of $\boldsymbol{\Lambda}$ ordered as in $\boldsymbol{\theta}_i$ (backward Letter, Gabor, and Pi; spatial Letter, Gabor, and Pi; OSPAN, CSPAN, SSPAN, and UCMRT) and the columns ordered as in $\boldsymbol{\eta}_i$,

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_1^V & \lambda_1^M & 0 & \lambda_1^S & 0 & 0 & 0 \\ \lambda_2^V & \lambda_2^M & 0 & 0 & \lambda_2^S & 0 & 0 \\ \lambda_3^V & \lambda_3^M & 0 & 0 & 0 & \lambda_3^S & 0 \\ \lambda_4^V & 0 & \lambda_4^M & \lambda_4^S & 0 & 0 & 0 \\ \lambda_5^V & 0 & \lambda_5^M & 0 & \lambda_5^S & 0 & 0 \\ \lambda_6^V & 0 & \lambda_6^M & 0 & 0 & \lambda_6^S & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_7^C \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_8^C \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_9^C \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{10}^C \end{pmatrix}.$$

Each masking task thus loads on the general visual factor (λ_j^V), on the factor for its masking procedure (λ_j^M), and on the factor for its stimulus (λ_j^S), and each cognitive task $m = 1, \ldots, 4$ loads on the cognitive factor (λ_{6+m}^C). All other loadings are fixed at zero.

The structural model specifies the relations among the factors,

$$\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{\Psi})\,, \qquad \mathbf{\Psi} = \mathbf{I}_7,$$

where $\mathbf{B}$ is a 7×7 matrix of regression weights among the factors. The cognitive factor is regressed on all six non-cognitive factors:

$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b_V & b_{\text{Bwd}} & b_{\text{Sim}} & b_{\text{Let}} & b_{\text{Gab}} & b_{\text{Pi}} & 0 \end{pmatrix},$$

so that $\eta_i^C = b_V \eta_i^V + b_{\text{Bwd}} \eta_i^{\text{Bwd}} + b_{\text{Sim}} \eta_i^{\text{Sim}} + b_{\text{Let}} \eta_i^{\text{Let}} + b_{\text{Gab}} \eta_i^{\text{Gab}} + b_{\text{Pi}} \eta_i^{\text{Pi}} + \zeta_i^C$. Setting $\mathbf{\Psi} = \mathbf{I}_7$ fixes the variances of the six non-cognitive factors and of the residual ζ_i^C to one and makes the six non-cognitive factors mutually orthogonal. Because the non-cognitive factors are orthogonal and have unit variance, this structure separates the two questions raised above. First, the variance of each masking task decomposes additively,

$$\text{Var}(\theta_{ij}) = \left(\lambda_j^V\right)^2 + \left(\lambda_j^M\right)^2 + \left(\lambda_j^S\right)^2 + \delta_j^2,$$

so the proportion of variance due to each source is its term divided by $\text{Var}(\theta_{ij})$. Second, the relation of each non-cognitive factor $f \in \{V, \text{Bwd}, \text{Sim}, \text{Let}, \text{Gab}, \text{Pi}\}$ to cognitive performance is captured by its regression weight b_f, and the corresponding correlation r_f is

$$\text{Var}\left(\eta_i^C\right) = \sum_f b_f^2 + 1, \qquad r_f = \frac{b_f}{\sqrt{\text{Var}(\eta_i^C)}}.$$

Full details of the model specification are provided in Appendix B. Derivations of the variance decomposition are provided in Appendix C.

Sources of variance in masking performance. Table 1 shows how the variance of each masking task was distributed across the general visual factor, the masking factors,

the stimulus factors, and task-unique variance. The general visual factor was the largest shared source of variance. It accounted for 32% of the variance on average. The stimulus factors were the second largest shared source, accounting for 18% of the variance on average. The masking factors accounted for the smallest share, 11% of the variance on average. The remaining 40% of the variance was unique to each task.

The decomposition shows two results: first, masking-specific variance was small, accounting for roughly a third of the share of the general visual factor. Backward and spatial masking were thus largely held together by the general visual factor rather than forming distinct sources of individual differences. Second, stimulus-specific variance tended to be larger than masking-specific variance (posterior probability = .84), in line with the clustering by stimulus reported above.

Contributions to cognitive performance. Table 2 shows the correlation of each non-cognitive factor with cognitive performance. Bayes factors are Savage–Dickey density ratios. We interpreted them following Kass and Raftery (1995): BF_{10} values above 3 provide positive evidence for a relation, and values above 20 provide strong evidence. Only the general visual factor was reliably related to cognitive performance, $r = .58$, 95% CI [.22, .82], and the Bayes factor provided strong evidence for this relation, $BF_{10} = 34.1$.

None of the specific factors showed a comparable relation. All five credible intervals included zero, and the correlations did not share a common sign. All Bayes factors fell in the inconclusive range. Thus, the data provided no evidence that stimulus-specific or masking-specific variance contributes to cognitive performance.

Taken together, stimulus content accounted for a substantial share of individual differences in masking performance, yet this stimulus-specific variance showed no evidence of a relation to cognitive performance. The relation between masking performance and cognitive performance was instead carried by the general visual factor shared by all six masking tasks.

Discussion

We asked whether sensory discrimination and processing speed, as indexed by spatial and backward masking of the same stimuli, reflect separable sources of individual differences. Three findings emerged. First, backward and spatial masking were largely held together by a general visual factor, which was the largest shared source of variance in the masking tasks; variance specific to the masking procedure was small. Second, there was a clear stimulus effect: tasks that shared a stimulus correlated most strongly, clustered together, and shared stimulus-specific variance beyond the general factor. Third, despite this stimulus effect, only the general visual factor was related to cognitive performance, whereas neither stimulus-specific nor masking-specific variance showed evidence of such a relation. Thus, varying the masking procedure did not reveal distinct abilities, and varying the stimulus introduced substantial variance that was unrelated to cognitive performance.

Sensory discrimination and processing speed are largely not separable

Sensory discrimination and processing speed have been studied in two largely separate literatures, each of which has been taken to reveal a distinct elementary source of cognitive ability (Deary, 1994). Our design held the stimulus and the mask constant and varied only whether performance was limited by the time available to encode the target (backward masking) or by the spatial overlap of target and mask (spatial masking). Under these conditions, the two procedures shared most of their reliable variance through a single general factor, and the variance specific to each procedure was small. If sensory discrimination and processing speed were distinct abilities, the masking factors should have been substantial, however they were not.

A stimulus effect that does not reach cognitive performance

The second finding concerns the role of the stimulus. The novel finding is that this stimulus-specific variance showed no evidence of a relation to cognitive performance.

What does the general visual factor reflect?

Methodological implications

The present analysis combined a trial-level psychometric model with a structural equation model at the latent level. Thresholds were estimated directly from trial-level responses, so the correlations among tasks were not attenuated by trial-to-trial noise (Rouder & Mehrvarz, 2026). Placing the factor model at the latent level, rather than fitting it to a matrix of estimated correlations, extended the uncertainty of the thresholds into the variance decomposition and the regression weights. The same approach can be applied whenever abilities are measured with psychophysical thresholds.

Appendix A: Hierarchical Psychometric Model With an Unstructured Covariance Matrix

Model Specification. Let $i = 1, \ldots, I$ denote individuals, $j \in \{1, 2, 3, 4, 5, 6\}$ denote six masking tasks, and $k = 1, \ldots, K$ denote trials. Let y_{ijk} denote the binary correctness outcome on the k-th trial of individual i in masking task j. The data level model is then given by

$$y_{ijk} \sim \text{Bernoulli}(F(s_{ijk}; \theta_{ij}, \beta_j)) \,,$$

where the success probability is given by a task-specific psychometric function, modeled as a cumulative normal function of log stimulus strength with a lower asymptote at chance:

$$F(s_{ijk}; \theta_{ij}, \beta_j) = b_j + (1 - b_j)\, \Phi(\beta_j\, (s_{ijk} - \theta_{ij}))\,.$$

Here, $\Phi(\cdot)$ denotes the standard normal cumulative distribution function, and b_j is the guessing rate for task j, fixed at .5 for the two-alternative Gabor and Pi tasks and at 1/9 for the nine-alternative Letter tasks. The variable s_{ijk} denotes the stimulus strength on trial k for individual i on task j, which was log-transformed and then standardized across all trials and individuals of task j. Standardization places the six masking tasks on a common scale, so that the same priors can be used for all tasks; because it is a linear transformation within each task, it does not affect the correlations among tasks. The parameter θ_{ij} denotes the latent threshold of individual i on task j, expressed on the same scale as s_{ijk}. β_j denotes the slope of the psychometric function for task j.

At the latent level, each individual is characterized by a vector of ten person-level parameters: the six latent thresholds from the masking tasks and the standardized scores on the four cognitive tasks,

$$\boldsymbol{\theta}_i = (\theta_{i1}, \ldots, \theta_{i6}, c_{i1}, \ldots, c_{i4})^\top \,,$$

where c_{im} denotes the standardized score of individual i on cognitive task m. These ten parameters are assumed to follow a multivariate normal distribution

$$\boldsymbol{\theta}_i \sim \mathcal{N}_{10}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \,.$$

The cognitive scores enter the model as observed values of the last four elements of $\boldsymbol{\theta}_i$. Specifically, the observed standardized score x_{im} of individual i on cognitive task m is modeled as $x_{im} \sim \mathcal{N}(c_{im}, \tau^2)$ with $\tau = 0.01$, so that c_{im} is constrained to be essentially equal to the observed score.

Prior specification. Priors are needed for the task means $\boldsymbol{\mu}$, the slopes β_j, and the covariance matrix $\boldsymbol{\Sigma}$. The task means and slopes were given weakly informative normal priors, $\mu_t \sim \mathcal{N}(0, 5^2)$ for $t = 1, \ldots, 10$ and $\beta_j \sim \mathcal{N}(0, 5^2)$ for $j = 1, \ldots, 6$, which are wide relative to the standardized scale of s_{ijk} and of the cognitive scores. The covariance matrix was given a scaled inverse-Wishart prior (Huang & Wand, 2013). In this prior, $\boldsymbol{\Sigma}$ is decomposed into a vector of scale parameters and an unscaled covariance matrix, $\boldsymbol{\Sigma} = \mathbf{D}\,\boldsymbol{\Sigma}_0\,\mathbf{D}$ with $\mathbf{D} = \text{diag}(\sigma_1, \ldots, \sigma_{10})$, $\boldsymbol{\Sigma}_0 \sim \text{Inverse-Wishart}(\nu_0, \mathbf{I}_{10})$, $\nu_0 = 12$, and $\sigma_t \sim \text{Half-}t_3(0, 2.5)$.

Model Fitting. The model was implemented in Stan (Carpenter et al., 2017). We ran 4 chains, each with 10,000 tuning steps followed by 20,000 drawing steps. Convergence was assessed with the rank-normalized $\hat{R}$ statistic (Vehtari, Gelman, Simpson, Carpenter, & Bürkner, 2021), which was below 1.01 for all parameters.

Appendix B: Hierarchical Psychometric Model With Structural Equation Modeling

Model specification. The data level of this model is identical to that described in Appendix A: the trial-level outcomes y_{ijk} of the six masking tasks follow the psychometric function $F(s_{ijk}; \theta_{ij}, \beta_j)$. The two models differ only at the latent level. In Appendix A, the

person-level vector $\boldsymbol{\theta}_i$ followed a multivariate normal distribution with an unstructured covariance matrix $\boldsymbol{\Sigma}$. Here, $\boldsymbol{\theta}_i$ is generated by the measurement and structural models described in the main text,

$$\boldsymbol{\theta}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda}\,\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}_{10}(\mathbf{0}, \boldsymbol{\Delta})\,,$$
$$\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{I}_7)\,,$$

with $\boldsymbol{\Lambda}$ and $\mathbf{B}$ as given in the main text. The latent vector $\boldsymbol{\eta}_i$ contains the cognitive factor and six non-cognitive factors, namely the general visual factor, the two masking factors, and the three stimulus factors. Unlike in Appendix A, the cognitive scores are not tied to the last four elements of $\boldsymbol{\theta}_i$ through a small τ. Instead, these elements are the observed standardized scores themselves, and each is modeled through the measurement model as $c_{im} = \mu_{6+m} + \lambda^C_{6+m}\,\eta^C_i + \epsilon_{i,6+m}$, so that the variance of each cognitive task that is not shared with the cognitive factor is captured by its unique variance δ^2_{6+m}.

Prior specification. Priors are needed for the task means $\boldsymbol{\mu}$, the slopes β_j, the free factor loadings in $\boldsymbol{\Lambda}$, the unique variances in $\boldsymbol{\Delta}$, and the regression weights in $\mathbf{B}$. The task means and slopes were given the same priors as in Appendix A. Each free loading was given a half-normal prior to fix the sign of each factor, $\lambda^V_j, \lambda^M_j, \lambda^S_j, \lambda^C_{6+m} \sim \mathcal{N}^+(0, 1)$; all other elements of $\boldsymbol{\Lambda}$ were fixed at zero. Each unique standard deviation was given a half-t prior, $\delta_t \sim \text{Half-}t_3(0, 2.5)$. Each of the six free regression weights in $\mathbf{B}$ was given a standard normal prior, $b_f \sim \mathcal{N}(0, 1)$.

Bayes factors. The Bayes factor for the hypothesis that non-cognitive factor f is unrelated to cognitive performance compares a model in which b_f is free to a nested model in which $b_f = 0$, with all other parameters unchanged. Because the nested model is obtained by fixing a single parameter, the Bayes factor equals the Savage–Dickey density

ratio (Wagenmakers, Lodewyckx, Kuriyal, & Grasman, 2010),

$$\mathrm{BF}_{10} = \frac{p(b_f = 0)}{p(b_f = 0 \mid \text{data})},$$

where the numerator is the prior density at zero, $1/\sqrt{2\pi}$, and the denominator is the posterior density of b_f at zero, estimated from the posterior draws with a logspline density estimator.

Model Fitting. The model was implemented in JAGS (Plummer, 2003). We ran 4 chains, each with 10,000 tuning steps followed by 20,000 drawing steps. Convergence was assessed with the Gelman–Rubin $\hat{R}$ statistic (Gelman & Rubin, 1992), which was below 1.01 for all parameters.

Appendix C: Derivation of the Variance Decomposition

This appendix derives the two quantities reported in the main text: the proportion of variance of each masking task due to each source, and the correlation of each non-cognitive factor with the cognitive factor. Here and below, the index f runs over the six non-cognitive factors, namely the general visual factor, the two masking factors, and the three stimulus factors, $f \in \{V, \mathrm{Bwd}, \mathrm{Sim}, \mathrm{Let}, \mathrm{Gab}, \mathrm{Pi}\}$. Both quantities follow from the structural equation model specified in Appendix B. As a reminder, the structural model is given by

$$\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{I}_7)\,.$$

In the structural model, the cognitive factor is the only factor that receives paths from other factors. Each non-cognitive factor is therefore equal to its own residual, $\eta_i^f = \zeta_i^f$, whereas the cognitive factor is a weighted sum of the non-cognitive factors plus its own residual, $\eta_i^C = \sum_f b_f\,\eta_i^f + \zeta_i^C$.

The variance and covariance are therefore

$$\mathrm{Var}\left(\eta_i^f\right) = 1, \qquad \mathrm{Cov}\left(\eta_i^f, \eta_i^{f'}\right) = 0 \;\; (f \neq f'), \qquad \mathrm{Cov}\left(\eta_i^f, \eta_i^C\right) = b_f,$$
$$\mathrm{Var}\left(\eta_i^C\right) = \sum_f b_f^2 \; \mathrm{Var}\left(\eta_i^f\right) + \mathrm{Var}\left(\zeta_i^C\right) = \sum_f b_f^2 + 1.$$

Variance decomposition of the masking tasks. Each masking task j loads on the general visual factor and on two specific factors: the factor for its masking procedure, denoted $\mathrm{mask}(j) \in \{\mathrm{Bwd}, \mathrm{Sim}\}$, and the factor for its stimulus, denoted $\mathrm{stim}(j) \in \{\mathrm{Let}, \mathrm{Gab}, \mathrm{Pi}\}$; for example, $\mathrm{mask}(j) = \mathrm{Bwd}$ and $\mathrm{stim}(j) = \mathrm{Let}$ for the backward Letter task. The corresponding loadings are λ_j^V, λ_j^M, and λ_j^S, so that

$$\theta_{ij} = \mu_j + \lambda_j^V \eta_i^V + \lambda_j^M \eta_i^{\mathrm{mask}(j)} + \lambda_j^S \eta_i^{\mathrm{stim}(j)} + \epsilon_{ij}.$$

Because the unique term ϵ_{ij} is independent of the factors, the variance of θ_{ij} is

$$\begin{aligned}\mathrm{Var}(\theta_{ij}) = \; & \left(\lambda_j^V\right)^2 \mathrm{Var}\left(\eta_i^V\right) + \left(\lambda_j^M\right)^2 \mathrm{Var}\left(\eta_i^{\mathrm{mask}(j)}\right) + \left(\lambda_j^S\right)^2 \mathrm{Var}\left(\eta_i^{\mathrm{stim}(j)}\right) \\ & + 2\lambda_j^V \lambda_j^M \,\mathrm{Cov}\left(\eta_i^V, \eta_i^{\mathrm{mask}(j)}\right) + 2\lambda_j^V \lambda_j^S \,\mathrm{Cov}\left(\eta_i^V, \eta_i^{\mathrm{stim}(j)}\right) + 2\lambda_j^M \lambda_j^S \,\mathrm{Cov}\left(\eta_i^{\mathrm{mask}(j)}, \eta_i^{\mathrm{stim}(j)}\right) + \delta_j^2.\end{aligned}$$

Given that the covariance among non-cognitive factors is 0, and the variance of each non-cognitive factor is fixed to 1, the above equation can be simplified as

$$\mathrm{Var}(\theta_{ij}) = \left(\lambda_j^V\right)^2 + \left(\lambda_j^M\right)^2 + \left(\lambda_j^S\right)^2 + \delta_j^2.$$

Dividing each term by the total variance therefore gives the proportion of variance due to each source.

Contribution of each non-cognitive factor to cognitive performance. Given $\mathrm{Var}(\eta_i^C) = \sum_f b_f^2 + 1$ and $\mathrm{Cov}(\eta_i^f, \eta_i^C) = b_f$, the correlation between non-cognitive factor f

and the cognitive factor is

$$r_f = \frac{\mathrm{Cov}\left(\eta_i^f, \eta_i^C\right)}{\sqrt{\mathrm{Var}\left(\eta_i^f\right)\mathrm{Var}(\eta_i^C)}} = \frac{b_f}{\sqrt{\sum_{f'} b_{f'}^2 + 1}},$$

which was computed for each posterior draw; Table 2 reports its posterior mean and 95% credible interval.

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Ethical Approval: Something

Consent to Participate: Something

Consent for Publication: Something

Data Availability: Something

Code Availability: All data and code required to reproduce the analyses are available at .

Author Contributions: Something

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process:

During the preparation of this work the authors used Claude and ChatGPT in order to check the grammar and spelling, as well as for verifying some mathematical calculations. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Raw material

This finding was first demonstrated by Galton (1883). For Victorian scholars working at the height of Western colonialism, the finding made complete sense as intelligence was conceptualized as a unidimensional quantity (Gould, 1981). For Galton, the founder of statistical treatment of eugenics, individuals varied on a single dimension, intelligence, that encompasses all mental abilities. Hence, the correlation between low-level perceptual abilities and high-level cognition is fully expected.

The modern view of intelligence, developed after the horrors of World War II, is that intelligence is multifaceted. The emphasis is on understanding a diversity of human abilities, capabilities, and intelligences (Carroll, 1993; Schneider & McGrew, 2018; Sternberg, 1985). Here primary mental abilities are treated as multiple and measured separately. People are not described by a single ability but by a profile of abilities. Modern views of cognition fit well within this multifacted view of intelligence. Cognition is viewed as information processing, and key questions are how information is represented and operated upon (Jackendoff, 1987; Marr, 1982; Neisser, 1967). The building blocks of cognition are information channels (Pollack, 1952; Shannon, 1948), filters (Broadbent, 1958), small memory systems that have limited domains, capacities, and time scales (Atkinson & Shiffrin, 1968; Miller, 1956; Sperling, 1960), and mechanical processes such as accumulation, integration, gating, differencing, and racing (Townsend & Ashby, 1983). More recent building blocks include multiple reasoning systems (Kahneman, 2011), distributed representations (McClelland & Rumelhart, 1981), and feedback projections in processing (Bar, 2009; Parr, Mirza, Cagnan, & Friston, 2019). The link from these mechanisms to individual variation is straightforward—individuals have different abilities because there is variability in the efficiency and usage of these mechanisms. It is within this this more modern context that the link between low-level perceptual performance and high-level cognitive performance is striking. If mental abilities are mediated by multiple representations, processes, and mechanisms, why the link?

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Table 1
Proportion of variance of each masking task due to each source.

Task	General	Masking	Stimulus	Unique
Backward Letter	.19 [.01, .46]	.14 [.00, .42]	.23 [.01, .53]	.44 [.21, .71]
Backward Gabor	.28 [.04, .57]	.14 [.00, .44]	.19 [.00, .52]	.39 [.16, .68]
Backward Pi	.38 [.11, .66]	.05 [.00, .22]	.18 [.00, .50]	.39 [.18, .65]
spatial Letter	.34 [.09, .60]	.10 [.00, .34]	.16 [.00, .40]	.41 [.21, .66]
spatial Gabor	.41 [.14, .65]	.06 [.00, .23]	.09 [.00, .29]	.45 [.27, .66]
spatial Pi	.33 [.05, .67]	.15 [.00, .49]	.20 [.00, .54]	.31 [.11, .60]
Average	**.32 [.20, .45]**	**.11 [.03, .20]**	**.18 [.08, .29]**	**.40 [.30, .51]**

Table 2
Correlation of each non-cognitive factor with cognitive performance.

Factor	r	BF_{10}
General visual factor	.58 [.22, .82]	**34.1**
Backward masking	-.06 [-.46, .38]	0.50
spatial masking	.21 [-.25, .61]	0.79
Letter	-.22 [-.55, .20]	0.86
Gabor	.31 [-.16, .68]	1.47
Pi	.10 [-.29, .48]	0.53

Note. r is the posterior mean of the correlation r_f between each non-cognitive factor and the cognitive factor, with 95% credible interval. Bayes factors are Savage–Dickey density ratios with prior $b_f \sim \mathcal{N}(0, 1)$. BF_{10} values above 3 provide positive evidence for a relation, and values above 20 provide strong evidence.

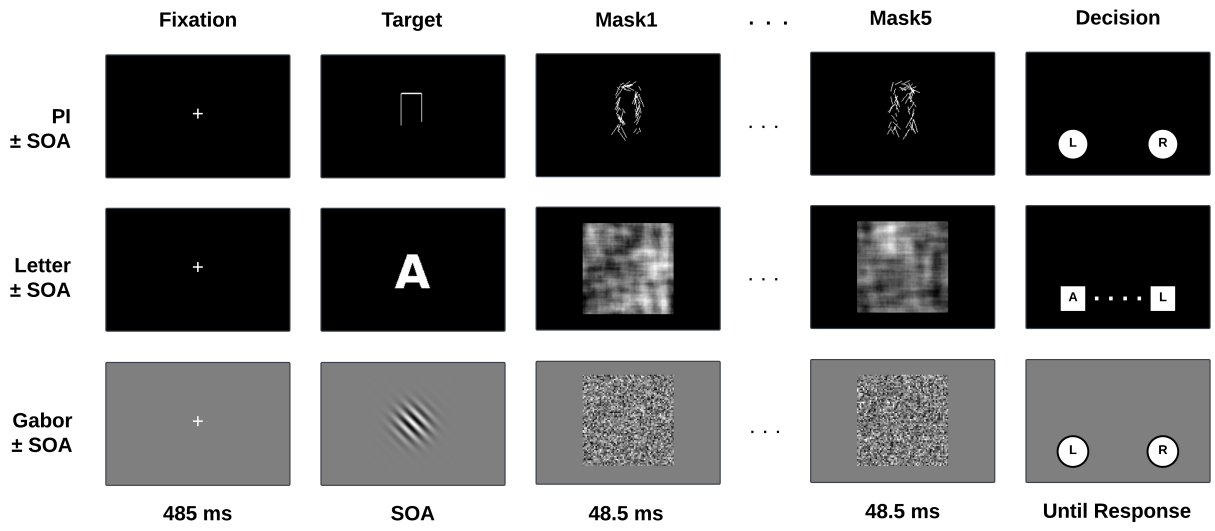


Figure 1. Backward visual masking tasks.

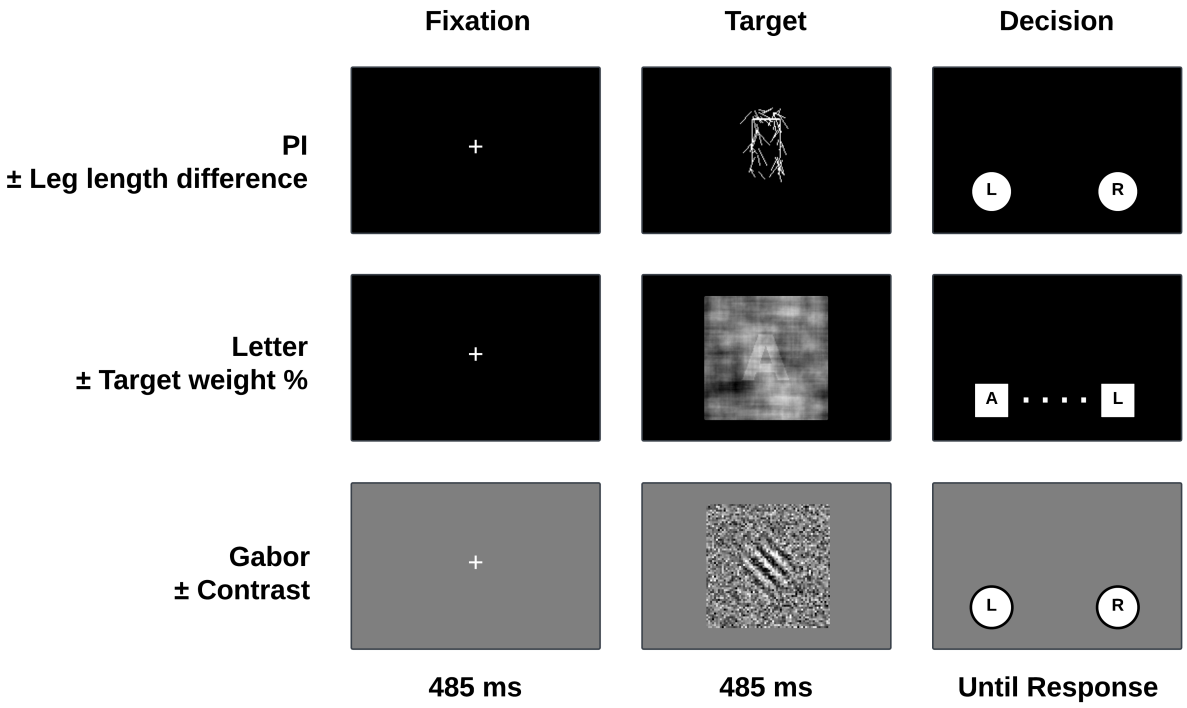


Figure 2. spatial visual masking tasks.

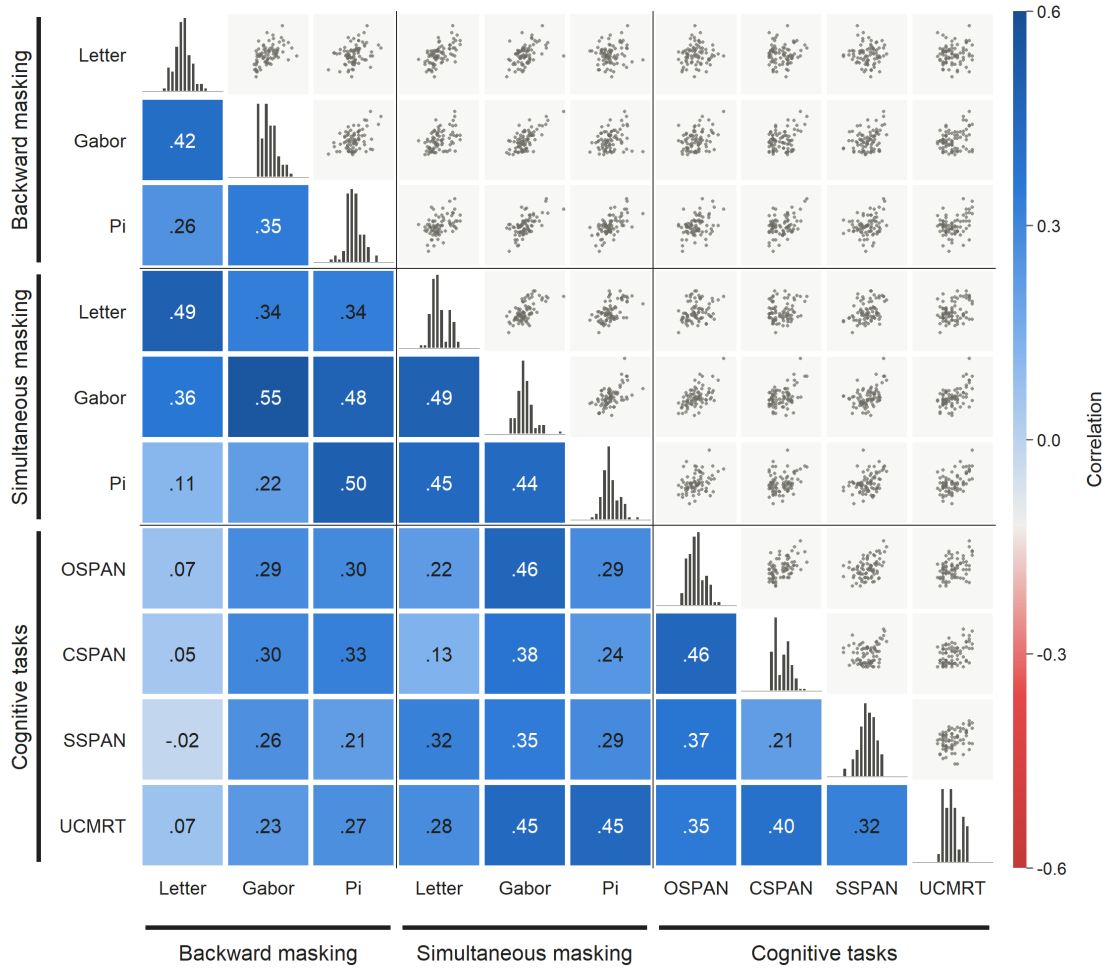


Figure 3. Model-estimated scores and correlations for the six masking tasks and the four cognitive tasks. Diagonal: distributions of the standardized scores. Lower triangle: model-estimated correlations. Upper triangle: scatterplots of the standardized scores.

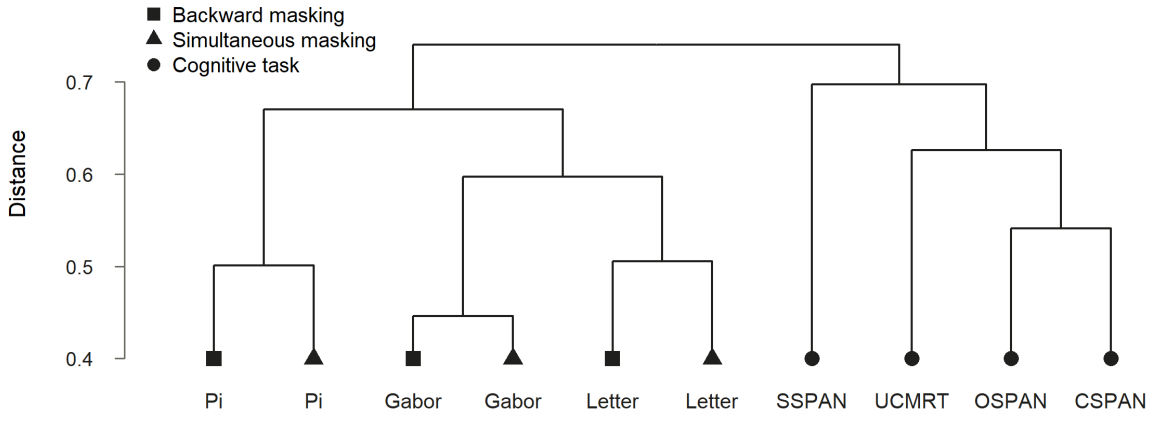


Figure 4. Hierarchical clustering of the ten tasks, based on the distance $1 - r$ between model-estimated correlations and average linkage. Symbols mark backward masking, spatial masking, and cognitive tasks.